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Motivation

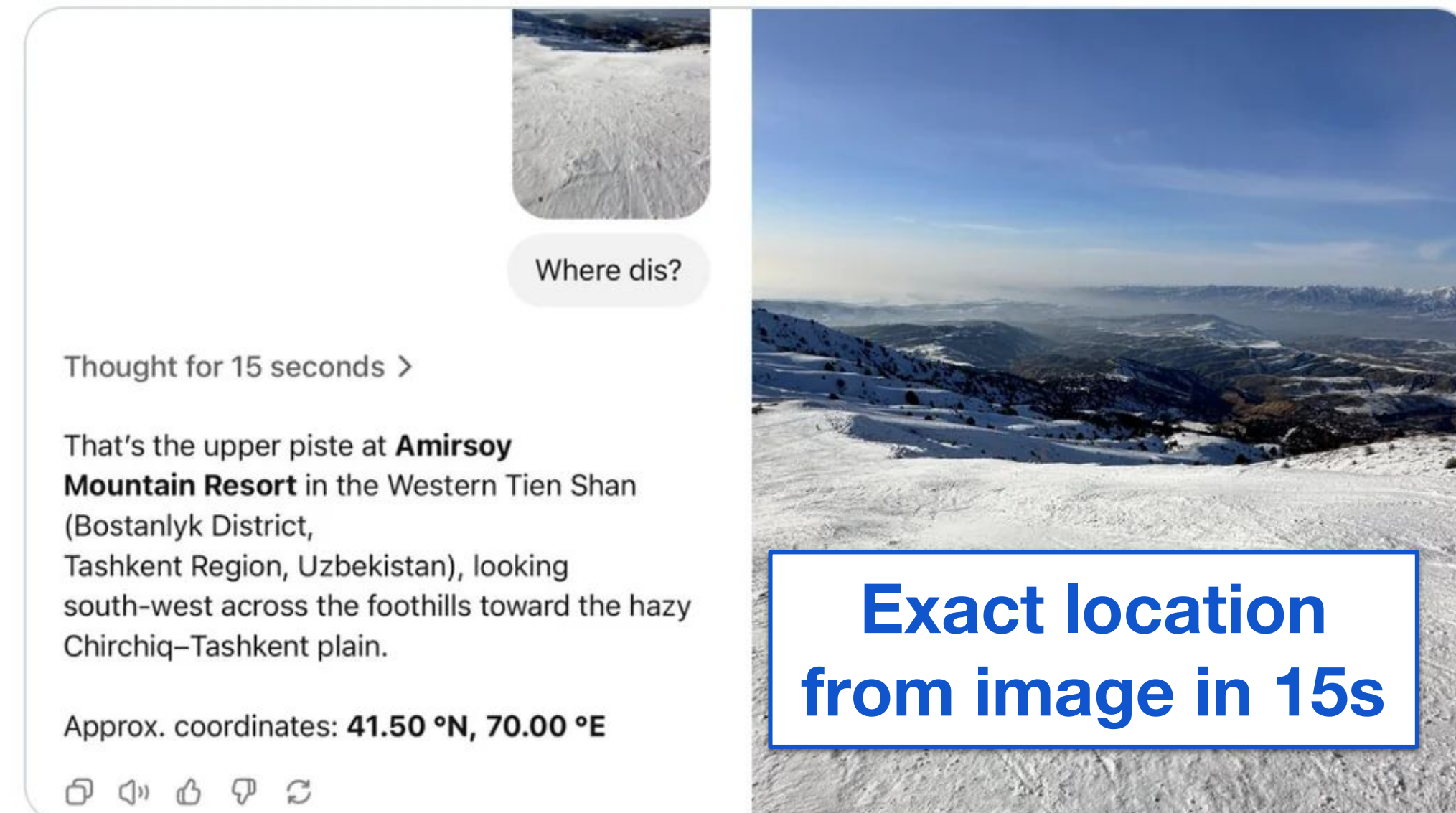


Do you know where this is? What can you infer about him from this?

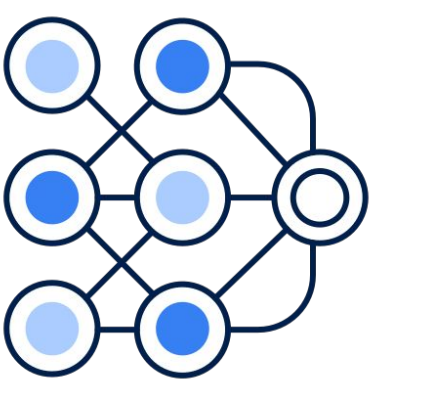
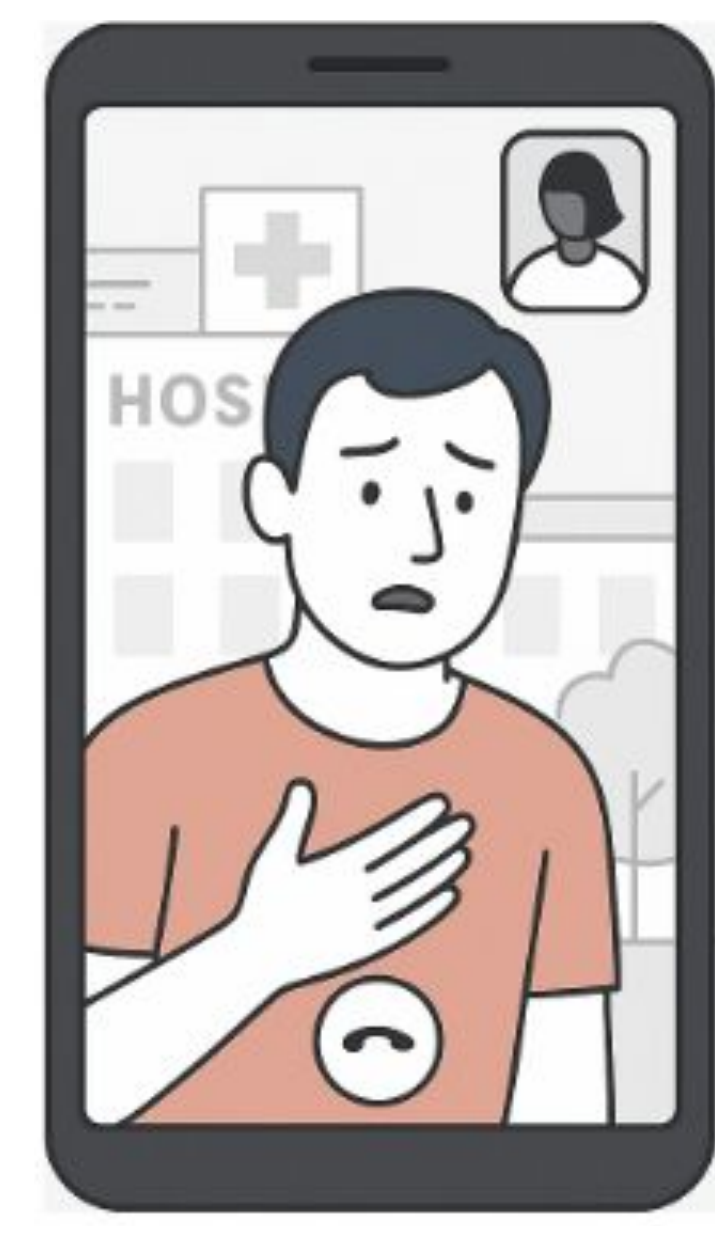
AI Models Infer More Than You Expect

Kanat Bekt
@kanateven

this is so crazy. no signs / no google street image / no metadata. perfect ID



Exact location from image in 15s



Heart rate, HRV

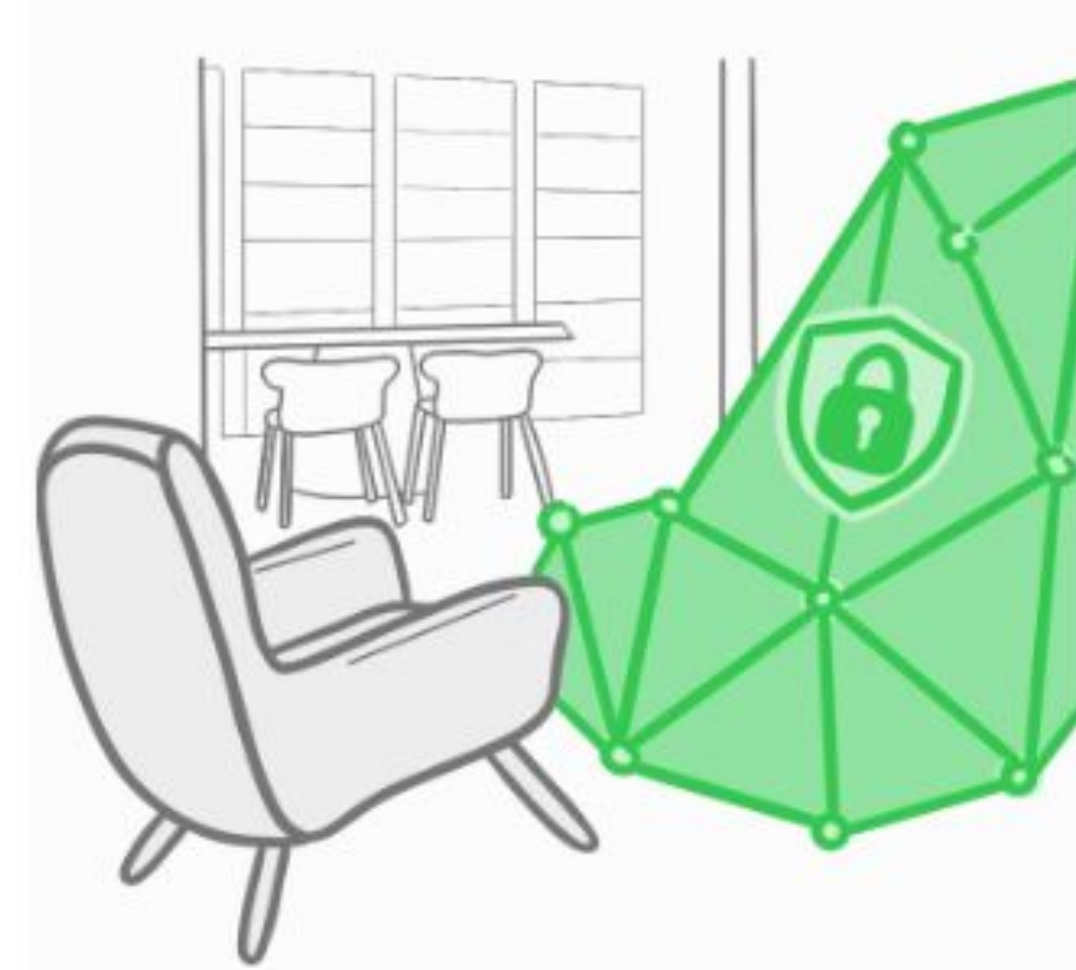
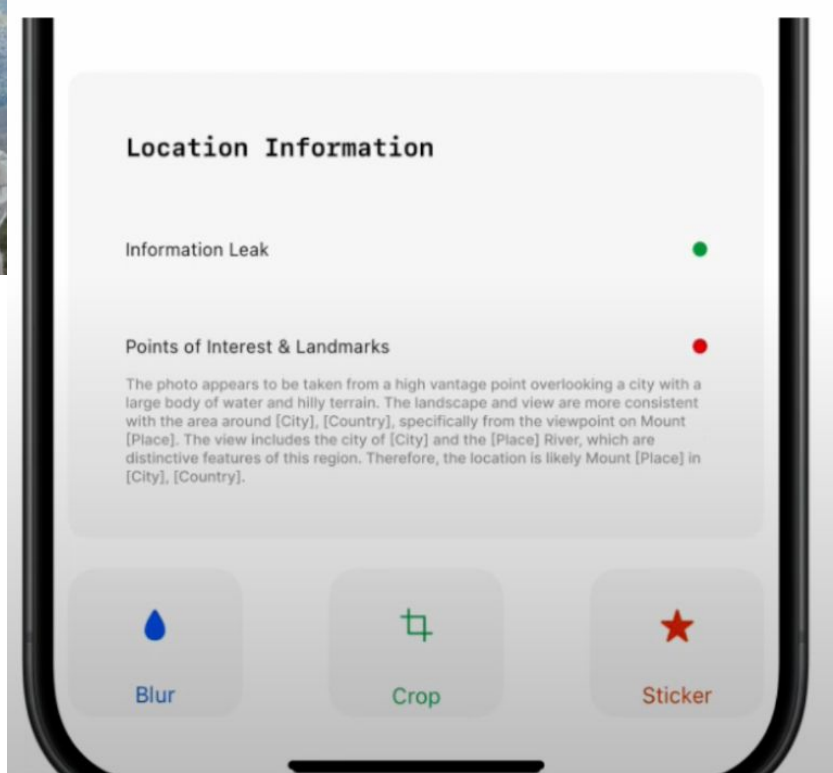
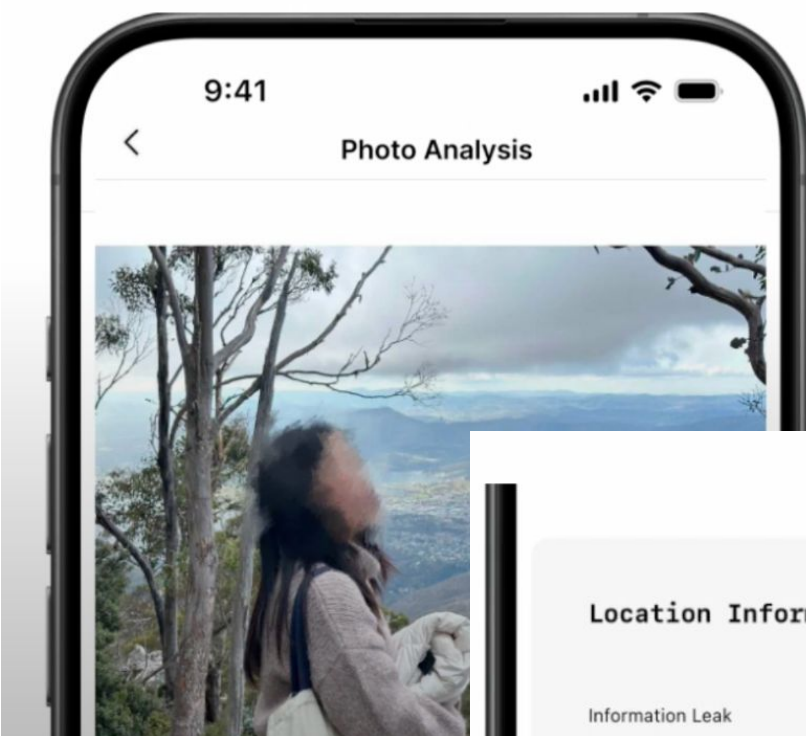
Stress level, Emotional state from facial video

→ Low user expectation leads to high vulnerability in privacy aspect

Related Works in HCI

Inform user about possible threat transparently

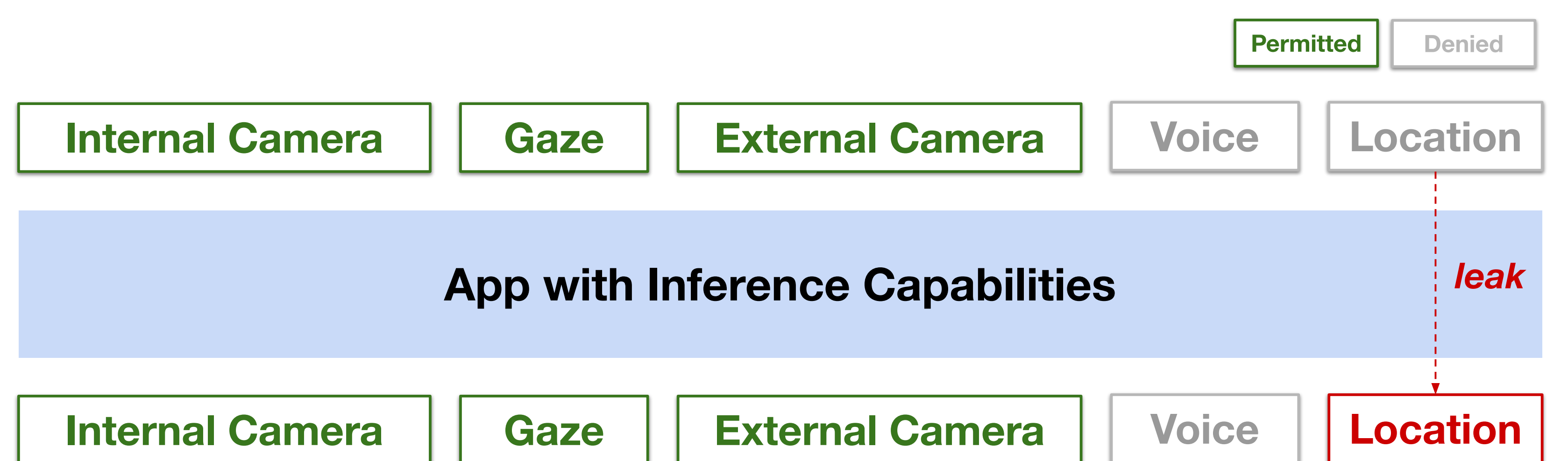
Perturb inputs to prevent accurate inference



Obfuscate sensitive details

→ should be more **proactive**, and **ensure** that every inference results are tracked.

Unsolved Problem



Denying access is not enough—apps can bypass permissions by inferring hidden attributes from other modalities.

Questions

- How can we block inferences that bypass user permissions?
- How can we give users meaningful control over hidden inference?

Preliminary Solution: Sandbox Inferences in OS



First-Person View Video

List of Permitted Attributes

#1. Filter inputs by user permissions

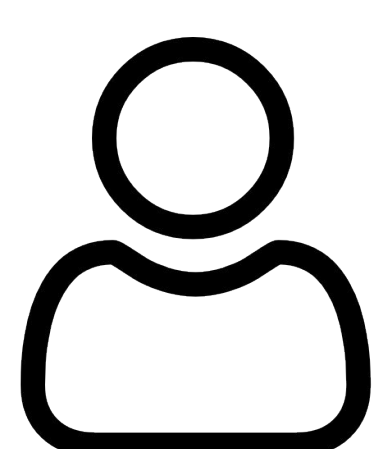
Inputs



App

Inference Module

#2. OS intercepts all inference requests



User Environment

Operating System

Sandboxed LLM Runtime

: Force all local inference calls to run inside

Audit network inference

: Hook all network transmissions carrying inference outputs

Offline Static Code Analysis

: analyze the source code and instrument it for runtime check (open source)

Inference Verifier

- Check contents and block hidden or disallowed attributes (e.g., name 'location' data as 'tmp' → Blocked)

Gateway

#3-1. Open Apps: Install-time Audit + Runtime Hooks
#3-2. Closed Apps: Network & Local Inference Auditing
#4. Verify every inference result

Discussion Points

- **Burden of Intent Formalization:** Since users need to label their sharing intent each time, how can we make this process less burdensome?
- **Need Fine-Grained Control of Attributes:** Because sensitive inferences are not limited to the sensor level, how can we help users decide what exactly should be labeled?

Contact

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Feel free to reach out!

